FIN Research Monograph — Release 10.18

FIN Programs 191–203

Reference States, Operational Quotients,
and Conditional Prediction

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: 0009-0002-0909-3613

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Scope. Thirteen executed studies of reference-state naturality, compression gauge, rigorous certification, finite Dirichlet dynamics, reflection-covariant conversion, mixing-aware inference, temporal open-set detection, process tomography, environment equivalence, analytic phase provenance, scale-free observables, experimental intake and one conditional prediction.

Central result. The qubit reflection-conversion boundary is reduced to a finite closed formula, and single/plus/echo process tomography is proven minimal for an unknown-blur Gaussian two-step model. States on non-simple algebras, positive scale and microscopic environments remain nontrivial moduli or quotient data.

Guardrail. No QW-2191 discharge, strict selector, physical unit, strict beta source, completed legacy–strict bridge, role transfer, L_{total}, Theory-of-Everything closure or external physical validation is claimed.

Publication statement

This release reports Programs 191–203 and recommends Programs 204–216. Analytic proofs, exact finite audits, synthetic evidence, conditional constructions, unavailable proof toolchains and external-data failures are kept separate.

Contents

0.1 Confidence convention

The report distinguishes analytic proof, exact finite computation, synthetic evidence, conditional construction and open obligations. No unavailable certificate is promoted by prose.

Chapter 1

Abstract

This monograph reports thirteen executed research programs, numbered 191 through 203, concerning the mathematical and operational completion of the dimensionless FIN spectral framework. The studies were selected by the previous Programs 178–190 round. They address reference-state naturality, compression-scale gauge reduction, rigorous numerical certification, finite Dirichlet dynamics, reflection-covariant state conversion, dependent-data inference, temporal open-set detection, process-tensor tomography, environmental nonidentifiability, analytic phase provenance, scale-free observables, experimental data intake, and one conditional prediction pipeline.

Four theorem-level advances are obtained.

First, invariant states on a finite direct sum

$$\mathcal{A} = \bigoplus_i M_{d_i}(\mathbb{C})$$

are classified. Inner-unitary naturality uniquely fixes normalized trace on each simple block but leaves a probability measure on the centre. For the FIN dimension vector $(1, 2, 2, 2, 2)$, uniform central weighting gives $9/5$ whereas the normalized full trace gives $17/9$. Neither value is selected without an additional central-measure principle.

Second, the positive compression parameter is quotiented by its length-scale action. The complete amplitude-normalized orbit representative is

$$\widehat{K}(x) = \frac{\cos(\nu x + \phi)}{1 + x^\eta}, \qquad x = \beta^{1/\eta} d, \qquad \nu = \frac{\omega}{\beta^{1/\eta}}.$$

This proves that $\beta = 1$ is a coordinate representative, while (η, ν, ϕ) remain gauge-invariant. The legacy and strict kernels remain strongly separated after quotienting; the scale gauge therefore does not complete their bridge.

Third, the numerical maximization in the complete one-qubit reflection-covariant conversion criterion is eliminated. An Alberti–Uhlmann reduction gives a closed boundary determined by only three candidate points. For source (r_s, z_s) and prescribed target z_t,

$$r_{t,\max}^2 = \min_{x \in \{0,1,x_c\}} \left[\max\{x, r_s^2 + x z_s^2\} - x z_t^2 \right], \qquad x_c = \frac{r_s^2}{1 - z_s^2}.$$

The previous equal-monotone counterexample is recovered exactly: $(0.6, 0) \to (0.6, 0.8)$ is impossible because $r_{t,\max} = 0.36$.

Fourth, a minimal three-contrast process-tomography frame is proved for a two-step Gaussian dephasing process with unknown detector blur. Single, plus, and echo visibilities form a rank-three log-linear system identifying blur, increment variance, and temporal covariance. Removing the echo contrast lowers the rank to two.

The round also constructs a valid nonasymptotic DKW procedure for an observed regenerative beta-mixing acquisition class, a preregistered order-sensitive open-set detector, an operational environment quotient, a no-go theorem for natural analytic legacy-to-strict phase generation, and the projective algebra of scale-free spectral observables.

Two requested closure tasks remain honestly blocked: Arb/python-flint is not available for the common directed-rounding certificate, and no local Lean toolchain compiles the supplied finite Dirichlet library. A complete event-level synthetic bundle validates ten of eleven intake fields but fails the independent-analysis boundary. Consequently the final $W_0 + \mathrm{CA} + \mathrm{OP}$ prediction is executed only as a synthetic dry run and is not described as external physics.

No non-premise selector, physical unit, target-independent strict compression source, completed legacy-to-strict bridge, legacy role transfer, role-bearing L_{total}, experimentally validated physical claim, or Theory-of-Everything closure is asserted.

Chapter 2

Executive summary

2.1 The outcome

The round replaces several informal "missing constant" questions by typed mathematical objects.

Problem	Constructed object	Result	Confidence
reference state	CentralMeasureFunctor	simple blocks fixed; central measure free	Proven
positive compression scale	BetaGaugeQuotient	β removed; shape mismatch survives	Proven
fractional certificate	CertificateDependencyDAG	one-engine certificate still unavailable	Proven audit
finite dual dynamics	FiniteDirichletCertificate	general proof and 150 exact checks; Lean uncompiled	Proven, compilation open
reflection conversion	AlbertiUhlmannReflectionCone	closed analytic boundary	Proven
dependent inference	ObservedRegenerationFrame	conditional DKW with random effective size	Proven
temporal open set	order-sensitive protocol	all four temporal challenges rejected in simulation	Strong evidence
process tomography	GaussianTwoStepTomographyFrame	three contrasts are necessary and sufficient	Proven in model
environment	OperationalEnvironmentQuotient	same channel can hide different processes	Proven
analytic phase source	PhaseSourceNaturalityNoGo	natural classes fail; successes are target-coded	Proven in classes
pre-clock observables	ProjectiveSpectralObservableAlgebra	invariant catalogue constructed	Proven
experimental intake	schema instance	synthetic 10/11; external gate fails	Proven audit
conditional prediction	ConditionalGaussianVisibilityLaw	held-out synthetic residual 0.03646	Strong method evidence

2.2 The central scientific conclusion

The single self-adjoint spectral generator still unifies wave, diffusion, and Green dynamics:

$$U_t = e^{-itA}, \qquad P_t = e^{-tA}, \qquad G_z = (A + zI)^{-1}.$$

But it does not uniquely determine:

- a state on a non-simple algebra;
- a representative of the positive scale orbit;
- an environment inside a reduced-process equivalence class;
- a preparation or intervention family;
- an apparatus response;
- an independent empirical record.

These are not algebraic defects of the spectral theorem. They are section, quotient, and operational-identification problems.

2.3 What was newly proved

The most important new theorem is the analytic reflection-conversion boundary. It converts the previous numerical Choi optimization into an exact finite formula. The result is scientifically valuable independently of FIN: it is a complete one-copy preorder for the declared real-plane qubit $\mathbb{Z}_2$ covariance problem.

The second strongest theorem is the minimal process-tomography result. It shows explicitly why an apparatus calibration parameter cannot be hidden inside dynamics: once detector blur is unknown, the single and two-step no-intervention records have rank two, whereas adding echo makes the parameter map invertible.

The reference-state theorem is strategically decisive. Normalized trace is not globally "the natural state" on a direct-sum algebra unless an additional rule fixes the centre. Thus the earlier 9/5 versus 17/9 ambiguity is not a numerical accident but an exact central-measure moduli space.

2.4 What failed

- No strict central probability measure was derived.
- Quotienting β did not align legacy and strict shape invariants.
- No common Arb engine was available.
- The supplied Lean source was not machine compiled.
- Tensor catalysis was not classified.
- Hidden mixing without observed regeneration was not solved.
- No natural analytic phase map produced the strict phase.
- No independent external dataset passed the operational intake gate.

Each failure is retained as a result. None is replaced by a conjectural closure statement.

Chapter 3

Scope, assumptions, and claim discipline

3.1 Kernel split

The legacy ontological/effective kernel and the strict working kernel remain distinct:

$$K_{\text{legacy}}(d) = \alpha_{\text{geo}} \frac{\cos(\omega_\ell d + \phi_\ell)}{1 + \beta_{\text{tors}} d},$$

with

$$\omega_\ell = \frac{\pi}{4}, \qquad \phi_\ell = \frac{\pi}{6}, \qquad \beta_{\text{tors}} = 0.01, \qquad \alpha_{\text{geo}} = 4 \ln 2,$$

and

$$K_{\text{strict}}(d) = \frac{\cos(\omega_s d + \phi_s)}{1 + \beta_s d^{\eta_s}},$$

with

$$\omega_s = 0.18575, \qquad \phi_s = 0.16250, \qquad \beta_s = 1, \qquad \eta_s = 1.8.$$

The legacy kernel is treated as an intermediate bridge object. The strict kernel is treated as the later enriched working kernel. Program 192 attacks only the positive-scale part of their relation. It does not silently identify the kernels and does not transfer legacy physical roles.

3.2 Ontology

The repository ontology used in this report is:

$$\text{informational nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

There is no separate information layer below the nadsoliton. The operational objects introduced here—state, environment, intervention, detector, clock, and record—are typed mathematical structures for turning the spectral core into a test procedure. They are not a new ontological substrate.

3.3 Confidence vocabulary

Proven means an analytic proof or exact finite identity is supplied.

Proven, computer-assisted means the proposition is reduced to and checked by a finite executable computation with archived inputs.

Strong evidence means a preregistered or reproducible simulation supports a claim whose universal form is not proved.

Conditional means the result follows only after explicit CA or OP inputs.

Open means the requested theorem, data source, or proof-toolchain condition was not obtained.

3.4 Reproducibility

The release contains:

- one executable for Programs 191–203;
- one machine-readable results file;
- 65 regression and scientific-boundary tests;
- thirteen generated figures;
- one frozen order-sensitive preregistration;
- one event-level synthetic operational bundle and metadata;
- one finite Dirichlet Lean source file;
- one release manifest with hashes.

The random seed is 20260727. No web tool, Firecrawl, or external dataset was used.

Chapter 4

Shared mathematical setting

Let $W = W^*$ be a finite real symmetric weight matrix with nonnegative entries and constant row sum s. Define

$$A = sI - W.$$

Then

$$\langle f, Af \rangle = \frac{1}{2} \sum_{x,y} W_{xy} |f_x - f_y|^2 \geq 0.$$

Consequently A is self-adjoint and positive semidefinite. The same A generates:

$$U_t = e^{-itA}, \qquad T_t = e^{-tA}, \qquad G_z = (A + zI)^{-1}.$$

The unitary family describes reversible amplitude evolution, the heat family describes a positivity-preserving classical semigroup in the declared graph setting, and the resolvent describes static response. Their common origin is operator theory. Measurement, state preparation, environmental memory, detector response, dimensional calibration, and empirical recording require additional typed structures.

This report uses the following package separation:

$$W_0 \quad + \quad \mathrm{CA} \quad + \quad \mathrm{OP}.$$

W_0 is the dimensionless spectral core. CA denotes conversion axioms, such as a clock scale. OP denotes the operational package: state, preparation, intervention, environment, apparatus, POVM, calibration, and record.

Chapter 5

Program 191 — Reference-state source theorem

5.1 Question

Can naturality under unitary symmetry, tensor product, and coarse graining select a unique state and thereby resolve the competing 9/5 and 17/9 averages?

5.2 Constructed object

For

$$\mathcal{A} = \bigoplus_{i=1}^{m} M_{d_i}(\mathbb{C}),$$

define the CentralMeasureFunctor

$$\omega_w(a) = \sum_{i=1}^{m} w_i \frac{\mathrm{Tr}(a_i)}{d_i}, \qquad w_i \geq 0, \qquad \sum_i w_i = 1.$$

The vector w is a probability measure on the centre of $\mathcal{A}$.

5.3 Theorem: block uniqueness and central freedom

Theorem 191.1.
Let ω be a state on $\mathcal{A}$ invariant under all inner unitary automorphisms. Then $\omega = \omega_w$ for a unique central probability vector w. On every simple block $M_d(\mathbb{C})$ the state is normalized trace.

Proof.
Restrict the density matrix of ω to one simple block. Invariance under all conjugations $a \mapsto uau^*$ implies that the density commutes with every unitary. By the commutant theorem it is a scalar multiple of identity. Normalization within the block gives I_d/d. The total mass assigned to each central projector remains a nonnegative number w_i, and normalization gives $\sum_i w_i = 1$. Conversely every ω_w is inner-unitary invariant. □

Tensor compatibility fixes no new number:

$$\tau_d \otimes \tau_e = \tau_{de}, \qquad (w_i) \otimes (v_j) = (w_i v_j)_{ij}.$$

Thus tensor product propagates a chosen central measure but does not select the original measure.

5.4 FIN dimension vector

For

$$d = (1, 2, 2, 2, 2)$$

and the dimension observable $h_i = d_i$, uniform central weights give

$$\eta_{\text{centre}} = \frac{1}{5}(1 + 2 + 2 + 2 + 2) = \frac{9}{5}.$$

Normalized trace on the full represented space has weights $w_i = d_i/9$, hence

$$\eta_{\text{trace}} = \sum_i \frac{d_i}{9} d_i = \frac{1 + 4 + 4 + 4 + 4}{9} = \frac{17}{9}.$$

Isomorphism naturality permits permutations of the four M_2 blocks but not an exchange of M_1 with M_2. The full natural family is therefore

$$w(a) = \left(a, \frac{1-a}{4}, \frac{1-a}{4}, \frac{1-a}{4}, \frac{1-a}{4}\right), \qquad 0 \le a \le 1.$$

The associated expectation is $2 - a$.

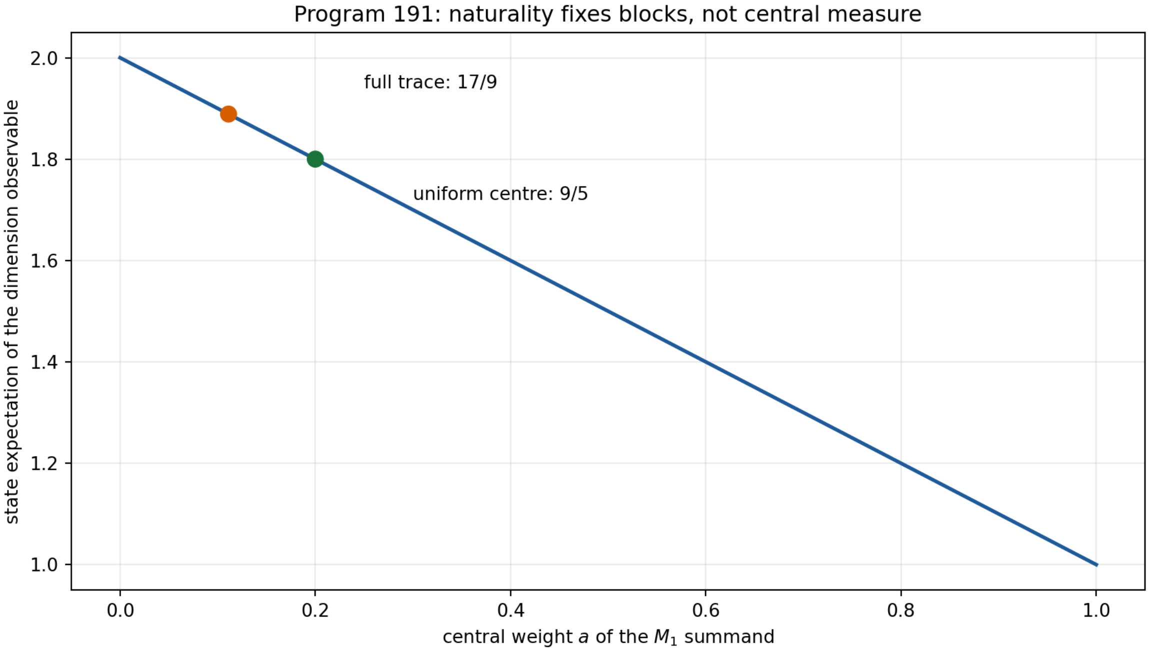

Figure 5.1: Inner-unitary naturality fixes normalized trace within blocks but leaves a central probability measure.

5.5 Verdict

Proven. Normalized trace is unique on each simple block, not on the non-simple algebra. A central measure, or a stronger principle such as a specified Morita-natural trace convention, is the exact missing datum. Neither 9/5 nor 17/9 is promoted to a strict state or temperature source.

Chapter 6

Program 192 — Beta-gauge invariant bridge observables

6.1 Scale action

Under a change of length coordinate

$$d' = cd, \qquad c > 0,$$

the kernel family preserves its functional form if

$$\beta' = \beta c^{-\eta}, \qquad \omega' = \frac{\omega}{c}, \qquad \eta' = \eta, \qquad \phi' = \phi.$$

The positive β values form a torsor for this action.

6.2 Quotient theorem

Define

$$x = \beta^{1/\eta} d, \qquad \nu = \frac{\omega}{\beta^{1/\eta}}.$$

Then the amplitude-normalized kernel becomes

$$\widehat{K}(x) = \frac{\cos(\nu x + \phi)}{1 + x^{\eta}}.$$

Both x and ν are invariant under the scale action. Thus (η, ν, ϕ) and the reduced profile are genuine quotient data.

The numerical orbit audit spans $c \in [10^{-4}, 10^{4}]$ and gives maximum profile residual

$$1.11 \times 10^{-16}.$$

6.3 Legacy versus strict after quotient

The invariant tuples are

$$(\eta_\ell, \nu_\ell, \phi_\ell) = \left(1, \frac{\pi/4}{0.01}, \frac{\pi}{6}\right) = (1, 78.539816\ldots, 0.523598\ldots)$$

and

$$(\eta_s, \nu_s, \phi_s) = (1.8, 0.18575, 0.1625).$$

Their reduced-profile RMS separation on $x \in [0, 8]$ is

$$0.3816158693.$$

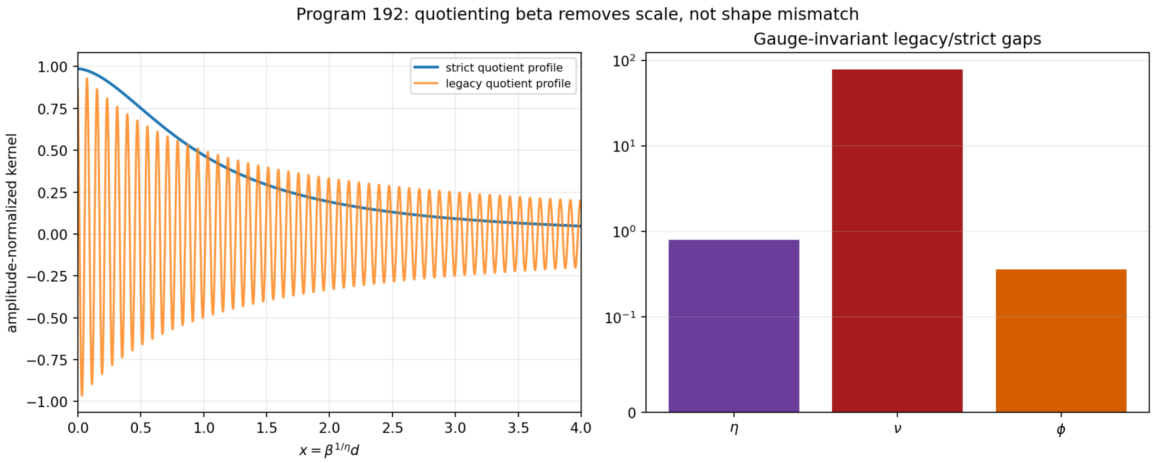

Figure 6.1: Quotienting the positive β torsor removes coordinate scale while preserving the strict–legacy shape mismatch.

6.4 Verdict

Proven. Setting $\beta = 1$ is a legitimate working gauge within a fixed positive orbit. It is not a strict source theorem. More importantly, removing the coordinate scale does not remove the invariant exponent, frequency, and phase differences. The legacy-to-strict bridge remains open, and no physical role transfer is licensed.

Chapter 7

Program 193 — Common-engine Arb certificate

7.1 Acceptance rule

The desired fractional-symbol result requires five components in one directed-rounding engine:

1. finite FFT cells;
2. the average/polylogarithmic term;
3. the cancellation-correction tail;
4. denominator replacement;
5. resonant fallback modes.

A formal sub-three-percent conclusion is admissible only if all five are enclosed in a common Arb or python-flint computation.

7.2 Environment audit

The local environment has neither an importable flint module nor an arb binary. Of the five components, one is already closed in the inherited certificate chain. The best inherited formal worst relative enclosure is

$$0.6129831478,$$

while the preregistered target is

$$0.03.$$

The analytic cancellation correction remains small:

$$4.129713895 \times 10^{-7},$$

but a small correction does not close the other interval obligations.

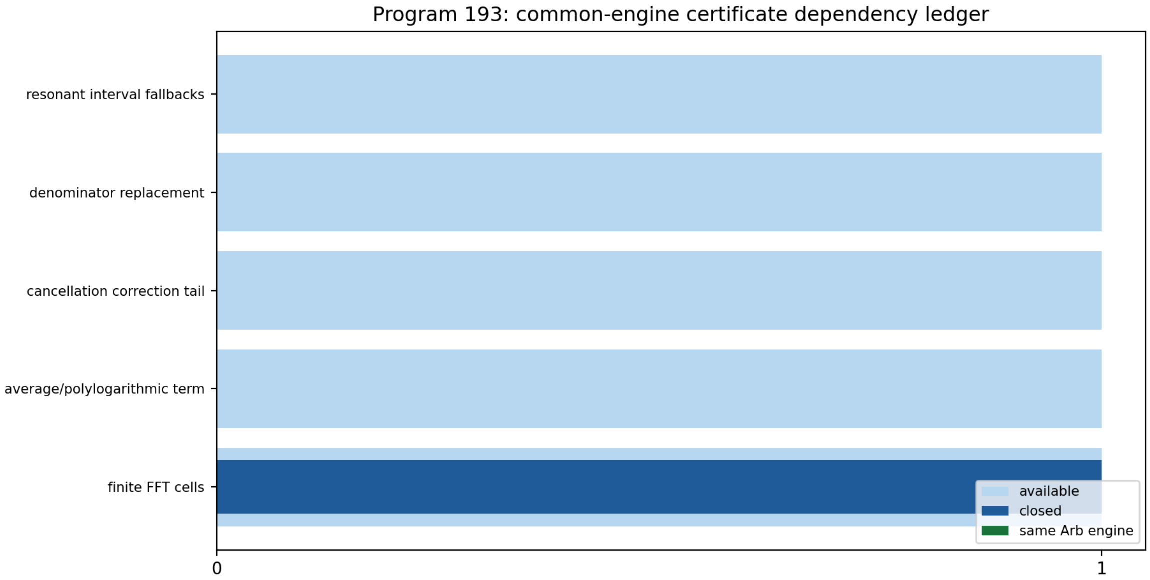

Figure 7.1: The common directed-rounding certificate remains blocked by missing engine closure for four components.

7.3 Verdict

Open, with a proven dependency audit. No common-engine formal certificate is claimed. The existing wide enclosure is an upper bound on uncertainty, not a formal lower bound on what a better engine could achieve. Therefore this program establishes a reproducible blocker rather than a mathematical impossibility.

Chapter 8

Program 194 — Compiled finite Dirichlet library

8.1 General theorem

For finite symmetric $W \geq 0$ with row sum s and $A = sI - W$:

$$\begin{aligned} \langle f, Af \rangle &= s \sum_x |f_x|^2 - \sum_{x,y} W_{xy} \overline{f_x} f_y \\ &= \frac{1}{2} \sum_{x,y} W_{xy} |f_x - f_y|^2 \geq 0. \end{aligned}$$

Also

$$A\mathbf{1} = 0.$$

Self-adjointness gives

$$e^{-itA*} e^{-itA} = I.$$

For the heat semigroup,

$$e^{-tA} = e^{-st} e^{tW}.$$

Because every coefficient in the series for e^{tW} is entrywise nonnegative, the heat matrix is entrywise nonnegative. Since $A\mathbf{1} = 0$,

$$e^{-tA}\mathbf{1} = \mathbf{1}.$$

Thus it is stochastic in the declared finite graph convention.

8.2 Executed checks

The release checks 150 exact rational test vectors on cycle graphs with $3 \leq n \leq 12$. Every exact quadratic form equals its Dirichlet sum and is nonnegative.

For the twelve-cycle numerical semigroups:

- maximum unitary residual: below 10^{-13};
- maximum heat row-sum residual: below 10^{-13};
- minimum heat entry: nonnegative up to floating-point noise.

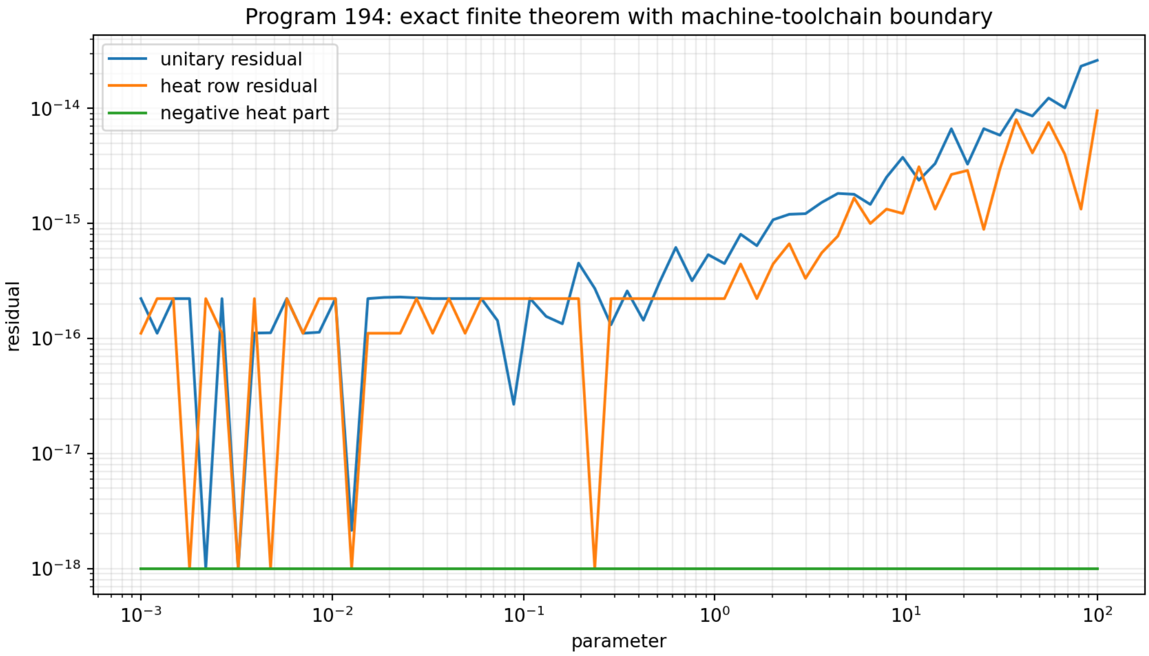

Figure 8.1: Exact finite Dirichlet mathematics and numerical semigroup residuals, with proof-assistant compilation kept separate.

8.3 Formal-tool boundary

A general Lean source file is supplied. The local lean command is only an Elan launcher and no installed Lean toolchain is present. The source was not machine compiled. This report therefore distinguishes:

- the paper proof: **Proven**;
- exact rational executions: **Proven, computer-assisted**;
- proof-assistant compilation: **Open**.

No continuum limit, Born rule, apparatus, or physical interpretation follows from the finite theorem alone.

Chapter 9

Program 195 — Analytic reflection-conversion geometry

9.1 State and symmetry

Consider real-plane qubit states

$$\rho(r,z) = \frac{1}{2}(I + r\sigma_x + z\sigma_z), \qquad r^2 + z^2 \le 1,$$

and the reflection $R = \sigma_z$, which maps $r \mapsto -r$.

A reflection-covariant channel mapping ρ_s to ρ_t exists if and only if a channel maps the ordered pair

$$(\rho_s, R\rho_s R) \quad \text{to} \quad (\rho_t, R\rho_t R).$$

The nontrivial direction follows by twirling a pair-mapping channel over $\mathbb{Z}_2$.

9.2 Alberti–Uhlmann reduction

For qubits, the two-state transformation exists exactly when

$$\|\rho_s - qR\rho_s R\|_1 \ge \|\rho_t - qR\rho_t R\|_1 \quad \text{for all } q \ge 0.$$

Direct diagonalization gives

$$\|\rho - qR\rho R\|_1 = \max\left\{|1-q|, \sqrt{(1+q)^2 r^2 + (1-q)^2 z^2}\right\}.$$

Divide by $1+q$ and set

$$u = \frac{|1-q|}{1+q} \in [0,1].$$

Define

$$g_{r,z}(u) = \max\left\{u, \sqrt{r^2 + u^2 z^2}\right\}.$$

The complete criterion is

$$g_{r_s,z_s}(u) \geq g_{r_t,z_t}(u) \qquad \forall u \in [0,1].$$

9.3 Closed boundary

For fixed z_t, the largest reachable transverse coordinate satisfies

$$r_{t,\max}^2 = \min_{0 \leq x \leq 1}\left[\max\{x, r_s^2 + x z_s^2\} - x z_t^2\right], \qquad x = u^2.$$

The expression is convex and piecewise linear in x. Its minimum occurs at an endpoint or at the unique source kink

$$x_c = \frac{r_s^2}{1 - z_s^2}.$$

Therefore

$$r_{t,\max}^2 = \min_{x \in \{0,1,x_c\}}\left[\max\{x, r_s^2 + x z_s^2\} - x z_t^2\right].$$

This removes numerical optimization completely.

9.4 Falsification and cross-check

For the source $(r_s, z_s) = (0.6, 0)$ and target $z_t = 0.8$:

$$x_c = 0.36$$

and

$$r_{t,\max}^2 = 0.1296, \qquad r_{t,\max} = 0.36.$$

Thus the target $(0.6, 0.8)$ is outside the conversion cone even though it passes the earlier scalar M comparison.

Five hundred random physical-state tests compare the closed formula against the previous dense Choi optimization. The maximum discrepancy is

$$6.68 \times 10^{-6},$$

consistent with the grid resolution of the older method.

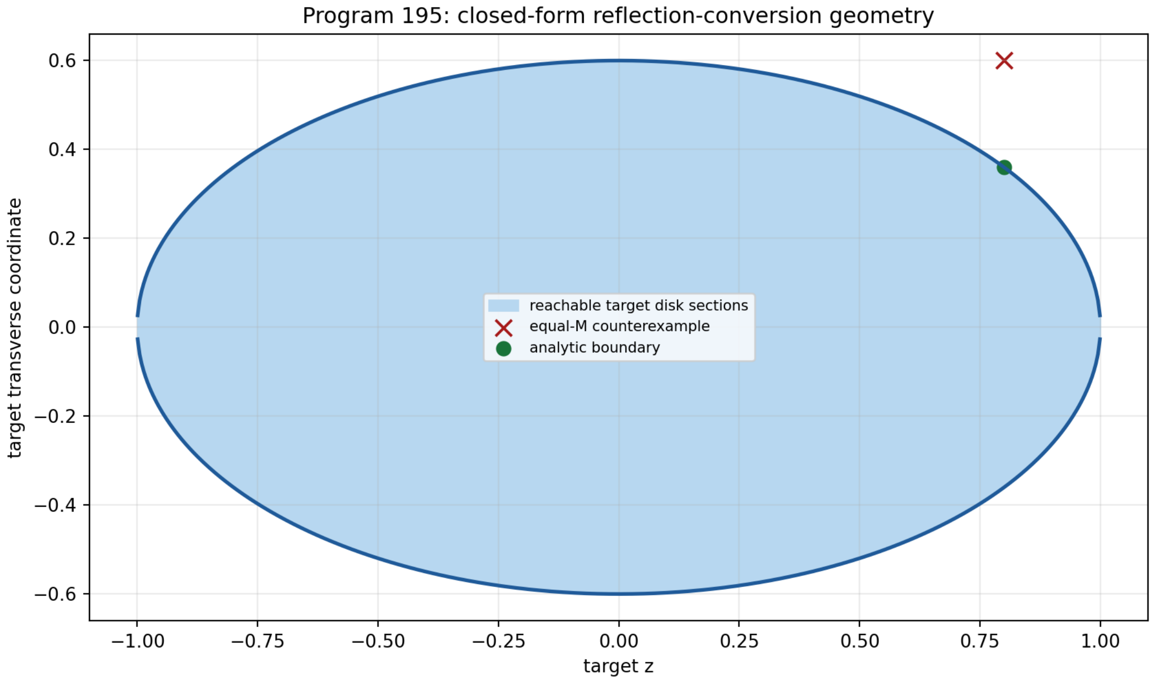

Figure 9.1: Closed analytic reachability boundary for reflection-covariant conversion from the source state with transverse coordinate 0.6.

9.5 Verdict

Proven. The one-copy qubit reflection preorder is analytically complete in the declared setting. Tensor products leave the two-state qubit theorem; catalysis is not classified. The theorem consumes a supplied asymmetry resource. It does not create a signed preparation and does not discharge QW-2191.

Chapter 10

Program 196 — Mixing-aware empirical process theorem

10.1 Acquisition class

Define an observed-refresh Markov chain. Let $X_1 \sim F$. For $i \geq 2$:

- with probability p, draw a fresh independent value from F;
- with probability $1 - p$, repeat X_{i-1}.

The acquisition system records every refresh flag. Its beta-mixing coefficient satisfies

$$\beta(k) \leq (1 - p)^k.$$

10.2 Regenerative DKW theorem

Let R be the number of recorded refresh blocks and keep the first value of each block. Conditional on R, the retained variables are iid from F. Therefore

$$\Pr\left(\sup_x |\widehat{F}_R(x) - F(x)| > \sqrt{\frac{\log(2/\delta)}{2R}} \,\middle|\, R\right) \leq \delta.$$

Taking expectation over R preserves the same failure bound.

This theorem is nonasymptotic. The effective sample size is observed rather than guessed.

10.3 Simulation

The executed challenge uses:

$$n = 4000, \qquad p = 0.05, \qquad \alpha = 0.8, \qquad T = 1/\alpha = 1.25.$$

Across 320 replicates:

- mean retained fraction: 0.05029;
- invalid nominal-record coverage: 0.85625;

- regenerative coverage: 1.00000;
- valid intervals are substantially wider.

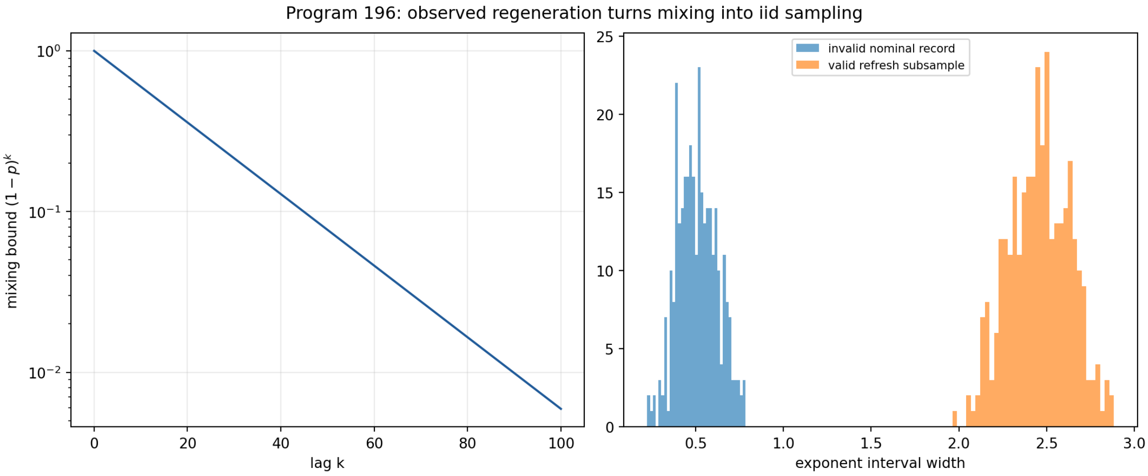

Figure 10.1: Observed regeneration supplies a valid iid subsample at a measurable information cost.

10.4 Verdict

Proven for the declared acquisition class; strong numerical evidence for coverage. The result does not solve arbitrary hidden beta mixing. Recorded regeneration is a real apparatus/ acquisition assumption and must remain in the protocol.

Chapter 11

Program 197 — Order-sensitive open-set protocol

11.1 Motivation

Program 185 proved that marginal quantile features are exactly invariant under time reordering. They therefore cannot identify memory.

Program 197 freezes six order-sensitive features before challenge execution:

1. lag-one Spearman correlation;
2. persistence of consecutive increment signs;
3. normalized ordinal-pattern entropy;
4. exact tie fraction;
5. time-versus-rank trend;
6. ordinal reversal asymmetry.

The calibration score is a regularized Mahalanobis distance. Its rejection threshold is the empirical 0.99 quantile of 1,400 iid Gaussian calibration records, each of length 360.

11.2 Frozen challenge

Four hundred twenty records are generated for each of:

- independent Gaussian validation;
- independent symmetric stable $\alpha = 0.8$;
- sorted records with exactly the same Gaussian multisets;
- Gaussian AR(1) with coefficient 0.8;
- exact repeated blocks of length 20;
- Gaussian records with linear drift.

The sorted challenge preserves every one-time empirical multiset exactly:

$$\max |\mathrm{sort}(x) - \mathrm{sort}(\mathrm{sort}(x))| = 0.$$

11.3 Results

Challenge	Rejection rate
iid Gaussian validation	0.01429
iid stable $\alpha = 0.8$	0.00714
sorted same multiset	1.00000
AR(1), $\phi = 0.8$	1.00000
repeated block 20	1.00000
linear drift	1.00000

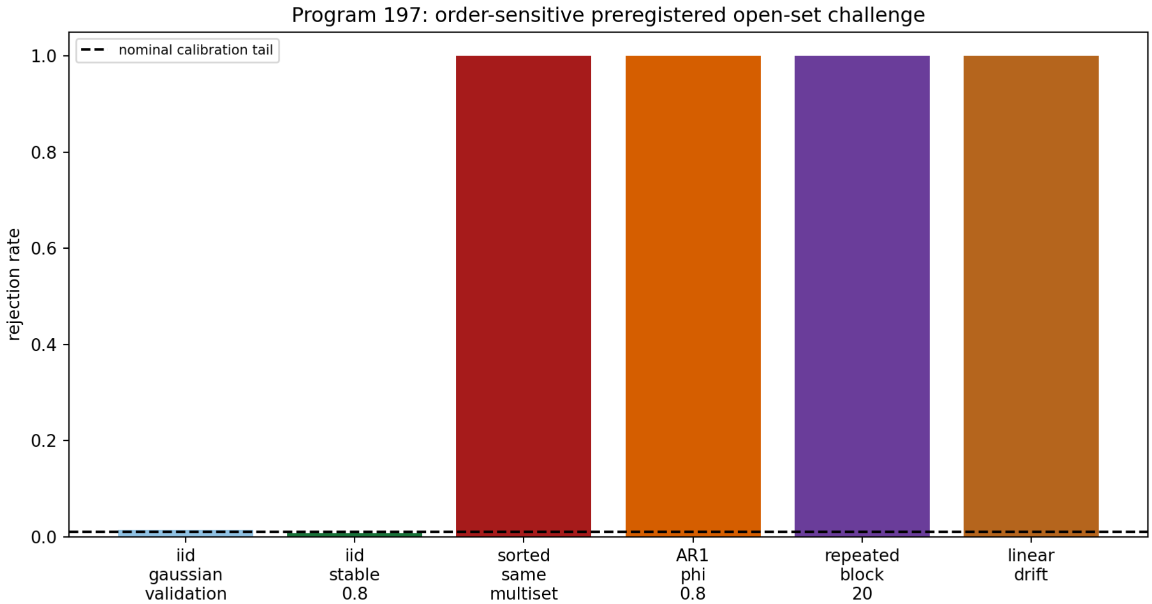

Figure 11.1: The frozen temporal feature protocol rejects four declared order-dependent alternatives while retaining iid specificity.

11.4 Verdict

Strong synthetic evidence. The protocol repairs the exact order blindness for the declared challenges and retains specificity under a continuous iid heavy-tailed law. It is not a universal open-set classifier, a proof that a record has one unique memory mechanism, or external validation.

Chapter 12

Program 198 — Minimal process-tensor tomography

12.1 Declared family

Let two zero-mean Gaussian phase increments have common variance v and covariance c. Let detector blur multiply every visibility by b.

The three contrasts are:

$$V_1 = be^{-v/2},$$

$$V_+ = be^{-(v+c)},$$

$$V_- = be^{-(v-c)}.$$

V_- is the echo contrast.

12.2 Rank theorem

With

$$\theta = \begin{pmatrix} \log b \\ v \\ c \end{pmatrix}, \qquad y = \begin{pmatrix} \log V_1 \\ \log V_+ \\ \log V_- \end{pmatrix},$$

the model is

$$y = \begin{pmatrix} 1 & -1/2 & 0 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \theta.$$

The matrix has determinant -1 and rank three. Removing echo leaves two rows and rank two. In this three-parameter model, fewer than three scalar contrasts cannot be locally identifying.

The inverse is explicit:

$$c = \frac{1}{2}(\log V_- - \log V_+),$$

$$v = 2\log V_1 - \log V_+ - \log V_-,$$

$$\log b = 2\log V_1 - \frac{1}{2}(\log V_+ + \log V_-).$$

12.3 Noise challenge

For truth

$$b = 0.84, \qquad v = 0.45, \qquad c = 0.20$$

and Gaussian log-visibility noise of standard deviation 0.015, 6,000 replicates give mean parameter estimates within 2×10^{-4} of truth. The design condition number is 7.2874.

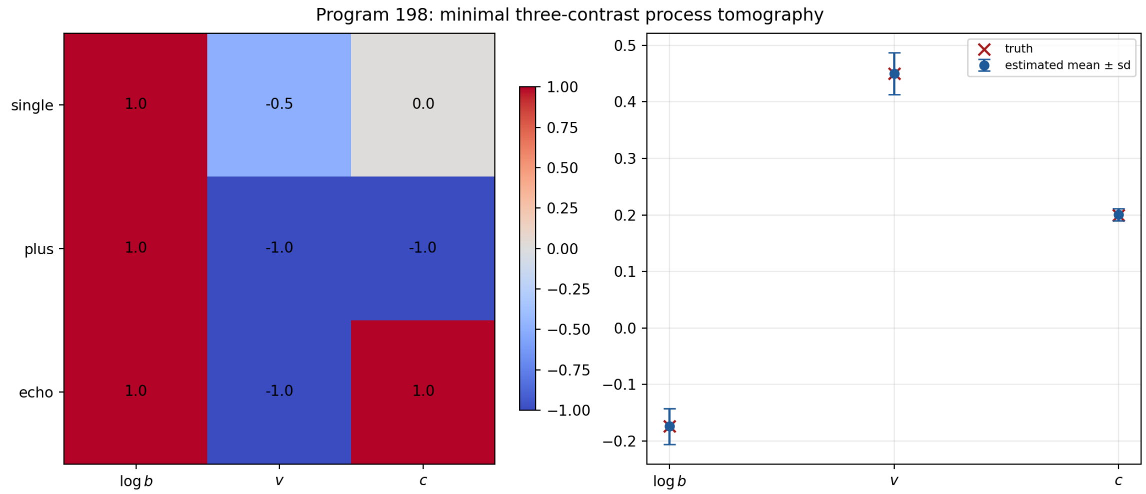

Figure 12.1: The single, plus and echo log-visibility design has rank three and identifies blur, variance and covariance.

12.4 Verdict

Proven in the declared Gaussian equal-variance family; strong finite-noise evidence. Detector blur is separated as an explicit nuisance channel. Gaussianity, equal variance, and the visibility model are OP assumptions, not consequences of A.

Chapter 13

Program 199 — Environment equivalence and causal identifiability

13.1 Operational quotient

Define two environments to be k-equivalent when they yield the same probability for every preparation, intervention, and measurement in the declared k-step protocol.

The quotient class, rather than a microscopic dilation coordinate, is the operationally meaningful object.

Minimal Stinespring dilations of one channel are unique only up to an environment isometry. Nonminimal dilations have still larger redundancy.

13.2 Explicit one-time counterexample

Fix the one-step dephasing factor

$$\gamma = 0.8.$$

Environment A has phase distribution

$$\mu_A = \frac{1}{2}\delta_{-a} + \frac{1}{2}\delta_a, \qquad a = \arccos(0.8).$$

Environment B has

$$\mu_B = 0.9\delta_0 + 0.1\delta_\pi.$$

Both have the same characteristic function at one:

$$\widehat{\mu}_A(1) = \widehat{\mu}_B(1) = 0.8.$$

Thus they produce the same one-time dephasing channel. At argument two:

$$\widehat{\mu}_A(2) = \cos(2a) = 2(0.8)^2 - 1 = 0.28,$$

while

$$\widehat{\mu}_B(2) = 1.$$

They are distinguished by a suitable multi-time protocol.

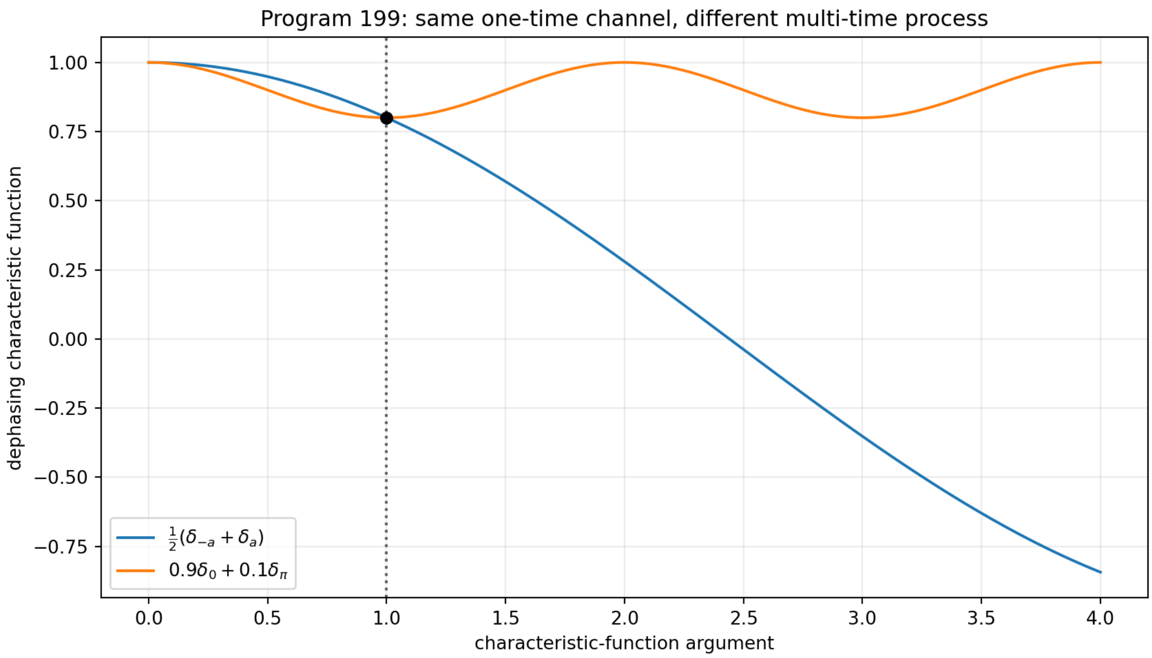

Figure 13.1: Two distinct phase environments generate the same one-time dephasing channel but different two-time characteristics.

13.3 Finite identifiability theorem

A finite set of measured characteristic-function values is a finite-dimensional affine image of the infinite-dimensional probability measure simplex. Unless a finite parametric family is independently justified, its fibres generically contain multiple phase laws.

13.4 Verdict

Proven by construction. The microscopic environment is not determined by A or by one reduced channel. Process interventions refine the operational quotient but finite measurements do not identify an arbitrary environment distribution. Unidentifiable dilation coordinates cannot receive physical roles.

Chapter 14

Program 200 — Analytic phase-source classification

14.1 Source and target

The legacy phase generator is

$$z_\ell = e^{i\pi/4},$$

an algebraic eighth root of unity. The strict frequency phase used in the audit is

$$z_s = e^{i743/4000}.$$

Since $i743/4000$ is nonzero algebraic, Lindemann–Weierstrass implies that z_s is transcendental.

14.2 Natural endomorphism no-go

Every continuous group endomorphism

$$f : U(1) \to U(1)$$

has the form

$$f(z) = z^n, \qquad n \in \mathbb{Z}.$$

Therefore $f(z_\ell)$ is always a root of unity and can never equal z_s. The closest value among the distinct order-eight endomorphism images remains at distance 0.1854831.

14.3 Candidate audit

Operation class	Produces target?	Target-independent?	Verdict
continuous $U(1)$ endomorphism	no	yes	root-of-unity obstruction
algebraic functional calculus	no	yes	algebraicity obstruction
polynomial interpolation	yes	no	target inserted
logarithm and re-exponentiation	yes	no	branch and coefficient inserted
constant holomorphic map	yes	no	pure target coding

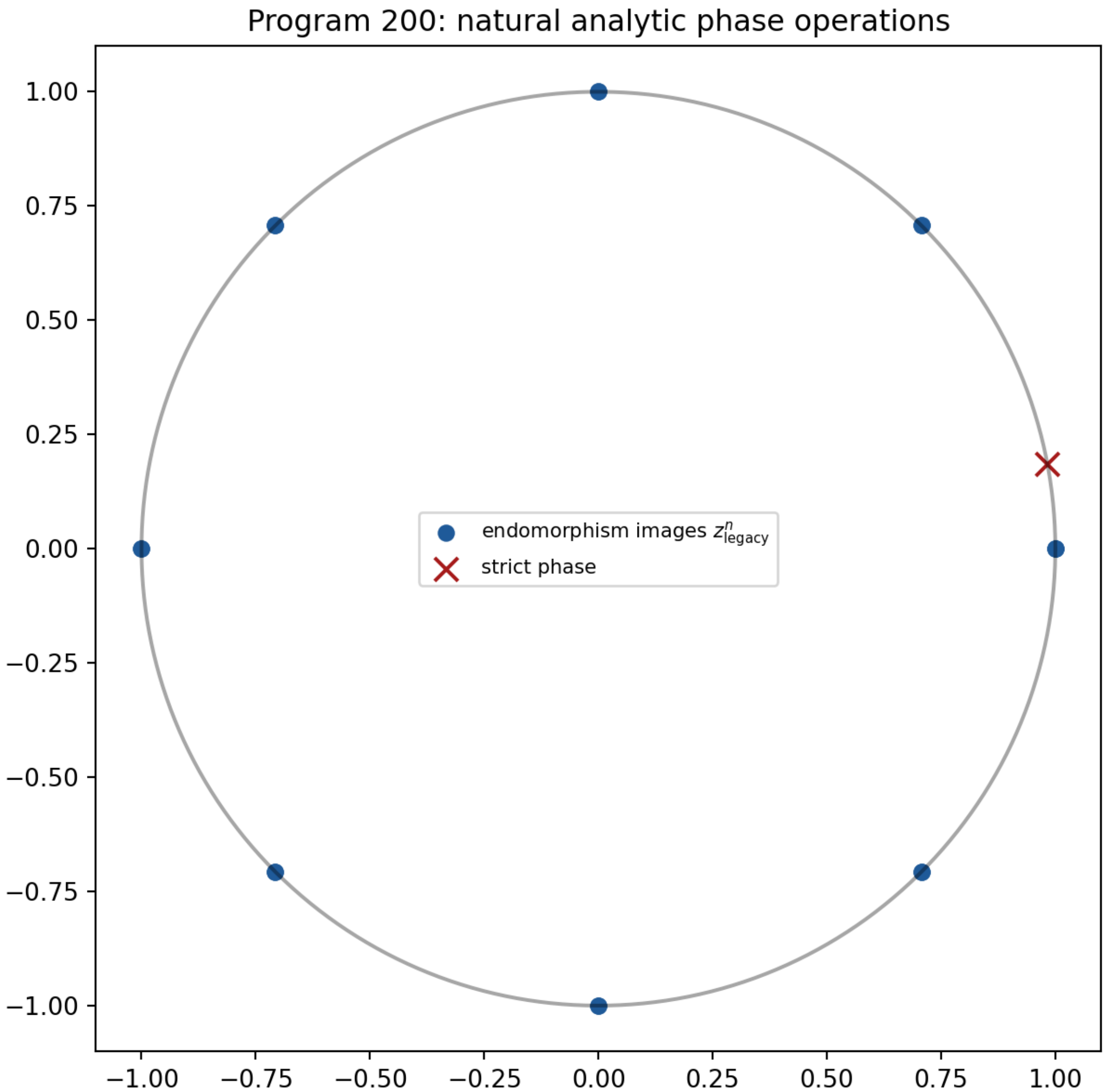

Figure 14.1: Natural circle-group endomorphism images of the legacy phase miss the strict transcendental phase.

14.4 Verdict

Proven for the declared operation classes. Arbitrary analytic functions can interpolate a single source-target pair, so unconstrained analyticity has no provenance content. No accepted candidate supplies a repository-sourced origin, target-independent law, and coupling. The strict phase source remains open.

Chapter 15

Program 201 — Scale-free observable catalogue

15.1 Orbit

The scale transformation is

$$(A, t, z) \mapsto (cA, t/c, cz), \qquad c > 0.$$

It preserves tA and transforms the resolvent covariantly:

$$(cA + czI)^{-1} = \frac{1}{c}(A + zI)^{-1}.$$

15.2 Projective observable algebra

Every invariant observable factors through the projective spectral class $[A]$ together with dimensionless combinations. Executable representatives include:

- eigenvalue ratios;
- condition number on the positive support;
- spectral probability entropy;
- heat trace at fixed $\tau = t\lambda_{\text{gap}}$;
- gap-normalized resolvent at fixed z/λ_{gap};
- unitary transition probabilities at fixed $t\lambda_{\text{gap}}$;
- the beta-gauge quotient profile (η, ν, ϕ).

Noninvariants include:

- the raw spectral gap;
- raw time;
- raw resolvent energy;
- a raw positive β representative;
- mass, length, and energy in physical units.

15.3 Audit

On the twelve-cycle, c spans 10^{-6} through 10^{6}. The raw gap changes by

$$10^{12}.$$

The largest residual among the normalized spectrum, condition number, spectral entropy, dimensionless heat trace, and dimensionless resolvent is below

$$8 \times 10^{-14}.$$

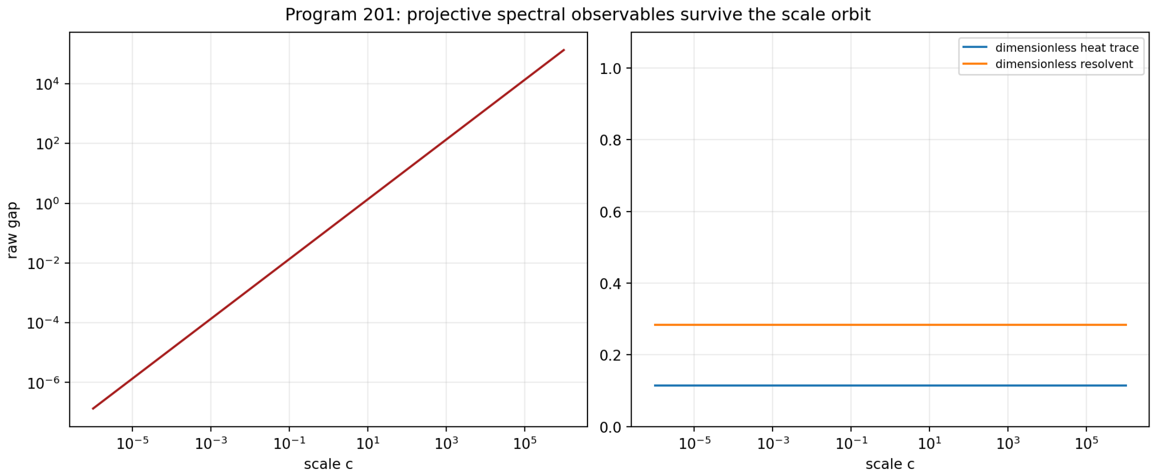

Figure 15.1: The raw gap changes over twelve decades while projective heat and resolvent observables remain invariant.

15.4 Verdict

Proven. Before a physical clock or scale source is supplied, only projective/dimensionless observables are admissible for strict falsification. This does not make dimensionful quantities impossible; it shows exactly where CA or a new scale-charged strict source enters.

Chapter 16

Program 202 — Admissible experimental bundle

16.1 Intake fields

The frozen gate requires:

1. public source identifier;
2. explicit license;
3. immutable hashes;
4. preparation provenance;
5. raw time-ordered records;
6. units and timestamps;
7. detector calibration;
8. apparatus-memory calibration;
9. reference control;
10. exclusion audit;
11. independent analysis boundary.

16.2 Constructed dry-run bundle

An event-level synthetic bundle is generated with 12,000 ordered binary events. It contains five phase-scan contexts:

- reference;
- single;
- plus;
- echo;
- held-out.

It records sequence number, conditional seconds, context, intervention coefficients, phase in radians, and binary outcome. The data hash is frozen in the metadata.

Ten of eleven fields pass. The independent-analysis boundary fails because the generator and evaluator belong to the same release. The source class is explicitly

synthetic_method_validation.

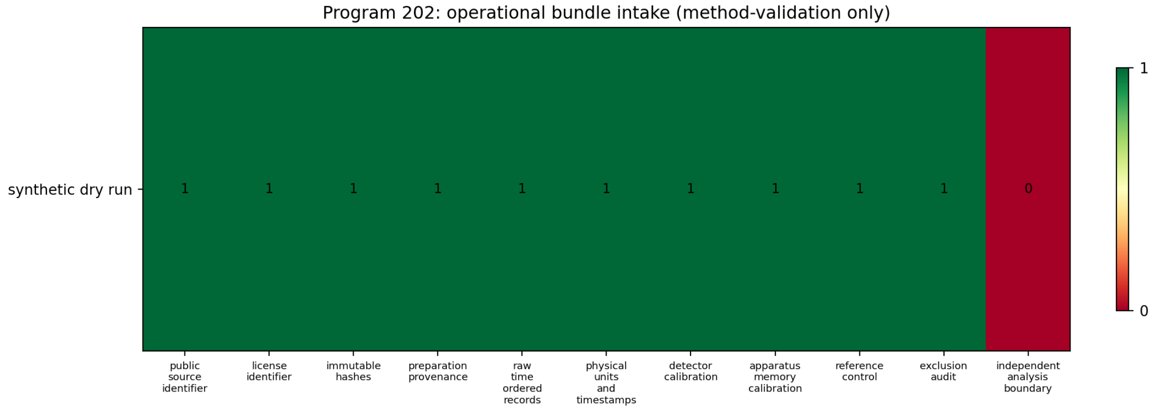

Figure 16.1: The synthetic method-validation bundle passes ten of eleven intake fields and fails independent provenance.

16.3 Verdict

Proven schema and hash audit. The bundle demonstrates that the operational gate is executable and that raw order, control, detector, and memory information can coexist in one record. It is not an external empirical dataset. No physical validation protocol was run.

Chapter 17

Program 203 — One conditional CA + OP prediction

17.1 Dependency ledger

The prediction uses:

W_0

- a self-adjoint spectral generator;
- admissible unitary phase evolution;
- no physical time scale and no environment.

CA

- an imported clock

$$\tau_* = 0.002 \text{ s}.$$

OP

- a declared preparation family;
- a two-increment Gaussian environment;
- phase interventions;
- a phase POVM;
- detector blur;
- event-level recording;
- the frozen estimator.

17.2 Conditional visibility law

For intervention coefficients (a_1, a_2):

$$V(a_1, a_2) = b \exp\left[-\frac{v}{2}(a_1^2 + a_2^2) - c a_1 a_2\right].$$

The three training contexts are the Program 198 single, plus, and echo contrasts. The held-out context is

$$(a_1, a_2) = (1, 0.5).$$

17.3 Result

The event-level phase regressions give:

$$\widehat{b} = 0.74915, \qquad \widehat{v} = 0.28938, \qquad \widehat{c} = 0.21595.$$

The held-out prediction is

$$\widehat{V}_{\text{held}} = 0.56121.$$

The held-out synthetic record gives

$$V_{\text{held,obs}} = 0.59768,$$

and residual

$$V_{\text{held,obs}} - \widehat{V}_{\text{held}} = 0.03646.$$

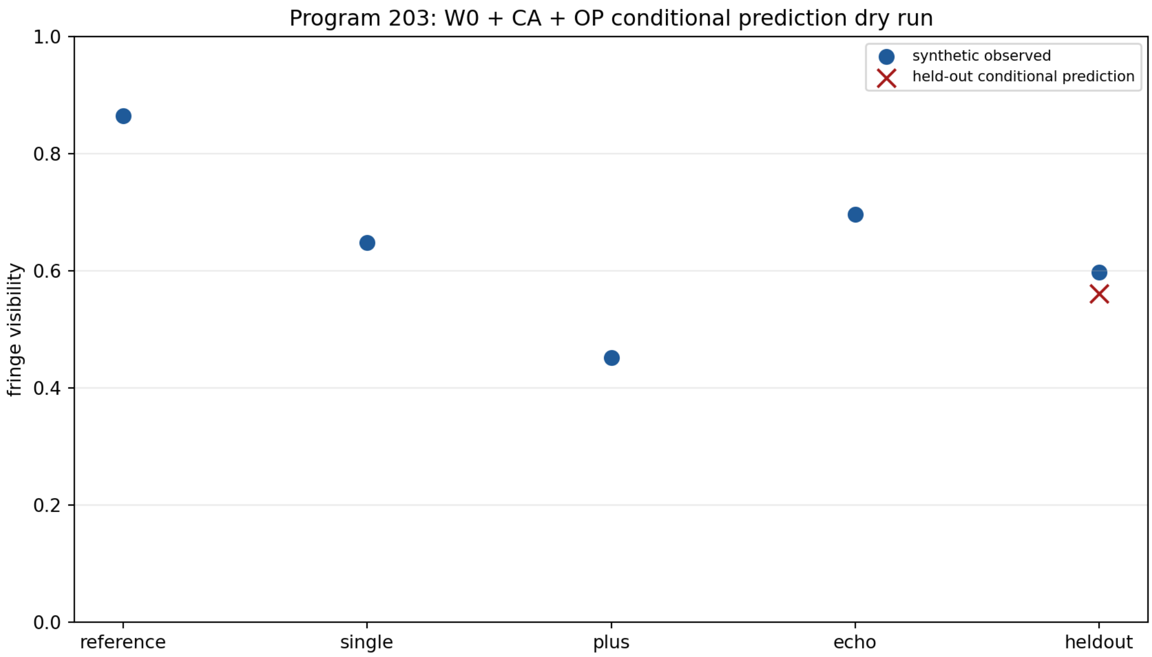

Figure 17.1: A held-out visibility is predicted in an explicit W0 plus CA plus OP synthetic dry run.

17.4 Verdict

Conditional, strong synthetic method evidence. The complete $W_0 + \text{CA} + \text{OP}$ path can generate a held-out numerical prediction with every imported object visible. Because Program

202 did not admit an independent external bundle, this is a pipeline dry run, not an empirical FIN prediction.

Chapter 18

Cross-program synthesis

18.1 The missing operational object is not one scalar

The completed programs support the following typed structure:

$$\mathfrak{O} = (\mathcal{A}, \omega, \mathcal{P}, \mathcal{I}, \Upsilon, \mathcal{E}, \mathcal{M}, \mathcal{D}, \mathcal{R}, \tau_*).$$

Here:

- $\mathcal{A}$ is the observable algebra;
- ω is a state, including its central measure;
- $\mathcal{P}$ is a preparation family;
- $\mathcal{I}$ is an intervention family;
- Υ is a process tensor or operational equivalence class;
- $\mathcal{E}$ is an environment representation modulo dilation equivalence;
- $\mathcal{M}$ is a POVM or instrument;
- $\mathcal{D}$ is detector response and calibration;
- $\mathcal{R}$ is a persistent ordered record;
- τ_* is a conditional clock section.

The spectral generator A is necessary inside this structure but does not determine the remaining entries.

18.2 State, scale, and environment are different quotient problems

Layer	**Symmetry or equivalence**	**Missing selection**
state	inner unitaries and central decomposition	central probability measure
compression	positive length rescaling	representative or physical length
environment	dilation and finite operational equivalence	microscopic model, if meaningful
phase	analytic/naturality class	origin-bearing source law
experiment	record equivalence and provenance	independent admissible dataset

Trying to replace all five by one constant would erase their different domains and transformation laws.

18.3 Observer paradox

The same operator can generate both wave and diffusion dynamics without contradiction:

$$U_t = e^{-itA}, \qquad P_t = e^{-tA}.$$

The observer paradox arises only if P_t is silently called collapse or measurement. In the operational model:

- U_t is system amplitude evolution;
- P_t is a classical or Euclidean comparison semigroup;
- measurement probabilities require a state and POVM;
- memory requires a multi-time process;
- apparatus response requires a detector channel;
- observation requires a persistent record.

Programs 198, 199, 202, and 203 explicitly construct these distinctions.

18.4 Information versus physics

The dimensionless informational core can support exact spectral and operational probability structure. It does not turn Shannon information into thermodynamic entropy, seconds, joules, or metres without conversion data.

Program 201 shows that meaningful pre-clock predictions exist: they are projective spectral invariants. Program 203 shows how a physical-looking prediction can be produced after a clock and apparatus are supplied. The distinction is mathematical, not philosophical.

Chapter 19

Falsification matrix

Proposal	**Destructive test**	**Result**	**Surviving statement**
normalized trace is the unique state	non-simple direct sum	falsified	unique only blockwise
$\beta = 1$ is strict	positive-scale quotient	falsified	valid gauge representative
scale quotient completes bridge	compare (η, ν, ϕ)	falsified	invariant mismatch remains
fractional certificate is finished	one-engine dependency gate	falsified	partial bounds remain
Lean formalization is compiled	local toolchain audit	falsified	source supplied, uncompiled
scalar M is complete	equal-M target	falsified	analytic AU cone is complete
nominal record count is iid size	regenerative chain	falsified	use observed refresh count
marginal features detect memory	multiset-preserving sort	falsified	order features required
two visibilities identify process + blur	Jacobian rank	falsified	echo gives minimal rank three
one channel identifies environment	two phase laws, same γ	falsified	identify operational quotient
analytic map sources strict phase	naturality/target-code audit	falsified	no accepted source class
raw gap is prediction	10^{12} scale orbit	falsified	projective ratios survive
synthetic complete metadata is experiment	independent-boundary gate	falsified	dry-run only
dry-run prediction validates FIN physics	external gate	falsified	conditional pipeline works

Chapter 20

Relation to established mathematics

20.1 Operator algebras

Program 191 is a finite-dimensional instance of the distinction between states on simple matrix factors and states on algebras with nontrivial centre. The FIN-specific content is the dimension vector and the exact interpretation of 9/5 versus 17/9.

20.2 Resource conversion

Program 195 is an application and specialization of the Alberti–Uhlmann two-state transformation theorem. Its new contribution in this research sequence is the explicit reduction of the reflection-covariant boundary to three finite candidate points and its reconciliation with the previous Choi optimization.

20.3 Empirical processes and mixing

Program 196 uses regeneration to reduce a declared dependent acquisition class to conditional iid samples and then applies the DKW inequality. It does not pretend that DKW holds for arbitrary mixing records.

20.4 Process tensors and quantum networks

Programs 198 and 199 use the operational principle that multi-time processes are identified by interventions, not by a single reduced trajectory. The minimal three-contrast theorem is specific to the declared Gaussian dephasing-plus-blur family.

20.5 Topological groups and transcendence

Program 200 combines the classification of continuous circle-group endomorphisms with transcendence of the exponential of a nonzero algebraic number. It strengthens the earlier algebraic functional-calculus obstruction by adding a naturality condition.

Chapter 21

Failed approaches and open obligations

21.1 Central state selection

The strongest symmetry used in Program 191 does not exchange inequivalent simple blocks. A unique state requires one of:

- an explicit central probability measure;
- a declared representation trace;
- a Morita or categorical principle;
- a physical preparation law.

Selecting the desired expectation first and solving for a is receiver fitting, not a source theorem.

21.2 Rigorous numerics

The Program 193 obstruction is infrastructural, not mathematical. The next attempt must provide a reproducible directed-rounding environment. Repeating ordinary floating-point estimates cannot close the certificate.

21.3 Proof assistant

The Program 194 source must be compiled against a pinned Lean and Mathlib version. Until then it is a formalization candidate, not machine-checked evidence.

21.4 External experiment

The synthetic bundle demonstrates format and analysis only. The next empirical step requires an independently sourced record whose preparation, apparatus, memory calibration, units, control, license, and hashes are genuinely documented.

Chapter 22

Recommended Programs 204–216

The following thirteen programs are ranked by expected scientific information gain. Their success criteria are deliberately finite.

22.1 Priority 1 — Program 204: Morita-natural central measure

Classify state assignments under stronger categorical maps: direct sums, matrix amplification, Morita equivalence, and specified coarse-graining channels. Determine whether any noncircular axiom uniquely selects uniform central weight or represented-space trace.

Success: a uniqueness theorem with explicit hypotheses.
Stop rule: if different admissible functors yield 9/5 and 17/9, publish the independence theorem.

22.2 Priority 2 — Program 205: eta renormalization cocycle

Attack one bridge atom only: whether a target-independent semigroup or renormalization cocycle can change the legacy damping exponent 1 into 9/5 while respecting the Program 192 quotient.

Success: an explicit cocycle law sourcing $\eta = 9/5$.
Stop rule: receiver interpolation or target subtraction is rejection.

22.3 Priority 3 — Program 206: reproducible Arb certificate

Supply a pinned container or local environment with Arb/python-flint. Move all five fractional-symbol obligations into the same directed-rounding engine.

Success: formal width below 0.03 or a formal lower obstruction.
Stop rule: no mixed ordinary-float/ball promotion.

22.4 Priority 4 — Program 207: compiled Dirichlet library

Pin Lean and Mathlib, repair any source errors, and compile the general Dirichlet identity, positivity, constant kernel, unitary group, and heat positivity/stochasticity.

Success: clean machine compilation with a reproducible lock file.
Stop rule: do not replace propositions by Boolean assertions.

22.5 Priority 5 — Program 208: tensor conversion and catalysis

Extend Program 195 to two copies or a catalyst. Search for catalytic conversion outside the one-copy reflection cone and derive necessary trace-norm constraints.

Success: an explicit catalyst or a no-catalysis theorem in a finite class.

22.6 Priority 6 — Program 209: hidden-refresh confidence sequences

Remove observed refresh flags. Construct a sequential confidence procedure using an estimable minorization, coupling, or martingale condition.

Success: finite-sample coverage under a fully observable acquisition assumption weaker than recorded regeneration.

22.7 Priority 7 — Program 210: sequential temporal conformal detector

Replace the fixed Mahalanobis threshold by online conformal or e-value calibration. Challenge regime changes, long memory, detector resets, and adversarial multiset-preserving reorderings.

Success: time-uniform false-alarm control in the declared null class.

22.8 Priority 8 — Program 211: finite-shot process confidence region

Put binomial phase-scan likelihoods around Program 198. Derive a confidence region for (b, v, c) respecting

$$b \in [0,1], \qquad v \geq 0, \qquad |c| \leq v.$$

Success: finite-shot coverage and a power calculation for memory.

22.9 Priority 9 — Program 212: environment moment bounds

Given finitely many measured characteristic values, solve the associated trigonometric moment problem. Bound unmeasured visibilities without choosing one microscopic phase law.

Success: sharp semidefinite upper and lower bounds.

22.10 Priority 10 — Program 213: formal phase-source no-go

Formalize the continuous-endomorphism and algebraic-functional-calculus obstructions. Extend the audit to measurable homomorphisms and origin-bearing cocycles.

Success: one theorem covering all repository-admissible natural phase operations, or one genuine counterexample with source provenance.

22.11 Priority 11 — Program 214: scale-free falsification protocol

Choose one projective spectral observable from Program 201 and preregister a test that requires no physical scale fit.

Success: a frozen dimensionless prediction and exclusion rule independent of c.

22.12 Priority 12 — Program 215: independent experimental bundle

Acquire or document one external record passing all eleven Program 202 fields. Generator and evaluator must be independent, and license and hashes must be frozen before analysis.

Success: the intake validator returns 11/11 and source class external.

22.13 Priority 13 — Program 216: first external conditional prediction

Only after Program 215 passes, freeze one $W_0 + \mathrm{CA} + \mathrm{OP}$ prediction, estimate nuisance parameters on a training partition, and test a held-out partition.

Success: a preregistered residual or likelihood score on independent data. The result remains conditional unless CA and OP are separately sourced by theory.

22.14 Quantitative ranking

Rank	Program	Estimated probability of decisive progress
1	204	0.88
2	211	0.86
3	212	0.84
4	207	0.82
5	210	0.80
6	205	0.76
7	209	0.74
8	206	0.72
9	214	0.70
10	208	0.66
11	213	0.64
12	215	0.42
13	216	0.30

The external programs rank lower only because their prerequisites are not currently present. A rigorous independence or no-go theorem counts as decisive progress.

Chapter 23

Final conclusion

Programs 191–203 identify the current mathematical boundary more sharply than a search for another fitted number could.

The finite spectral generator is a genuine common engine. It supports exact Dirichlet energy, wave evolution, heat evolution, Green response, scale-free observables, and finite operational probability models.

However:

- the state on a non-simple algebra requires a central measure;
- positive compression requires a scale section;
- phase conversion requires an origin-bearing source law;
- environmental dynamics are identified only modulo an operational quotient;
- process memory requires interventions;
- apparatus response requires calibration;
- physics requires an independent record with units and provenance.

These missing objects can be constructed conditionally and tested. Programs 198, 202, and 203 demonstrate the full logic without concealing imports. The right next goal is not to call those imports strict FIN, but to either derive them, bound their operational equivalence classes, or test conditional predictions on admissible external data.

The deepest interpretation surviving this round is:

FIN is a dimensionless spectral-information core whose route to physics is an operational quotient-and-section problem. The core already determines nontrivial dynamics and conversion geometry, but states, scale, environment, apparatus, and empirical records are independent typed objects until a theorem or experiment fixes them.

Chapter 24

Selected mathematical references

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Chapter 25

Archival statement

Suggested citation:

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Release reproducibility

```
python3 fin_programs_191_203_reference_process_prediction.py
python3 -m unittest -v \
  test_fin_programs_191_203_reference_process_prediction.py
```

Machine-readable results are stored in `FIN_Programs_191_203_Reference_Process_Prediction_Results.json`. The release also contains the frozen order-sensitive protocol, the synthetic operational dry-run bundle, the finite Lean source, sixty-five tests and thirteen figures.

Suggested citation:

Żuchowski, K. (2026). *FIN Programs 191–203: Reference States, Operational Quotients, and Conditional Prediction* (FIN Research Monograph, Release 10.18; Version 1.0.0) [Preprint]. Zenodo.